

CBSE Class 10 Mathematics

Time Allowed: 3 Hours

Maximum Marks: 75

General Instructions:

1. This question paper contains 30 questions. All questions are compulsory.
2. Questions are grouped into sections A through E.
3. Use of calculators is NOT permitted.

Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. If two positive integers a and b are written as $a = x^3y^2$ and $b = xy^3$, where x and y are prime numbers, then the result of $\frac{HCF(a,b)}{LCM(a,b)}$ is:
(A) (A) xy
(B) (B) x^2y^2
(C) (C) $\frac{1}{x^2y}$
(D) (D) $\frac{1}{xy}$
2. The decimal expansion of the rational number $\frac{147}{2^2 \times 5^7 \times 3}$ will terminate after how many places of decimals?
(A) (A) 2
(B) (B) 7
(C) (C) 5
(D) (D) 9
3. If the polynomial $p(x) = x^4 - 6x^3 + 16x^2 - 25x + 10$ is divided by $g(x) = x^2 - 2x + k$, the remainder is found to be $x + a$. What is the value of k ?
(A) (A) 2
(B) (B) 3
(C) (C) 5
(D) (D) 7
4. When $f(x) = 2x^3 + ax^2 + bx - 6$ is divided by $(x - 1)$ and $(x + 2)$, the remainders are 0 and 12 respectively. What is the value of $a - b$?
(A) (A) 8
(B) (B) 10
(C) (C) 12
(D) (D) 14

5. If the sum of the first n terms of an arithmetic progression is given by $S_n = 3n^2 + 5n$, and the k^{th} term of this progression is 164, then the value of k is:
- (A) (A) 25
 (B) (B) 26
 (C) (C) 27
 (D) (D) 28
6. Assertion (A): The number 0.121121112... is an irrational number. Reason (R): A number is irrational if its decimal expansion is non-terminating and non-repeating.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true
7. Assertion (A): The denominator of the rational number $17/80$ can be expressed in the form $2^n \times 5^m$, where n and m are non-negative integers. Reason (R): A rational number p/q has a terminating decimal expansion if the prime factorization of q is of the form $2^n \times 5^m$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true
8. Assertion (A): If a polynomial $p(x) = x^3 - 3x^2 + 5x - 3$ is divided by $g(x) = x^2 - 2$, then the degree of the remainder $r(x)$ is 1. Reason (R): According to the division algorithm for polynomials $p(x) = g(x)q(x) + r(x)$, the degree of the remainder must always be strictly less than the degree of the divisor $g(x)$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true
9. Assertion (A): If the polynomial $x^4 - 6x^3 + 16x^2 - 25x + 10$ is divided by $x^2 - 2x + k$, the remainder is $x + a$. If $k = 5$, then the value of a is -5 . Reason (R): The division algorithm states that the remainder $r(x)$ obtained by dividing $p(x)$ by $g(x)$ is unique.

- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
10. Assertion (A): If the sum of the first n terms of an AP is given by $S_n = 3n^2 + 5n$, then its common difference d is 6. Reason (R): The common difference of an AP can be calculated as $d = S_2 - 2S_1$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true

Section B: Very Short Answer (2 marks each)

1. If the *HCF* of 120 and 45 is expressed in the form $120m - 45n = 15$, find the value of $m + n$ where m and n are positive integers.
2. Show that the number $7 \times 11 \times 13 + 13$ is a composite number without performing the actual multiplication.
3. If the polynomial $p(x) = x^4 - 6x^3 + 16x^2 - 25x + 10$ is divided by $g(x) = x^2 - 2x + k$, the remainder is found to be of the form $x + a$. Find the values of k and a .
4. Given that $x^2 + 1$ is a factor of the polynomial $p(x) = x^4 + ax^3 + bx^2 + cx + d$, show that $b - d = 1$ and $a = c$.
5. If the n^{th} term of an Arithmetic Progression is given by $a_n = 5 - 3n$, find the common difference d and the sum of the first 10 terms, S_{10} .

Section C: Short Answer (3 marks each)

1. Show that for any positive integer n , the number 7^n can never end with the digit 0 or 5.
2. A merchant has three types of oil: 403 litres of mustard oil, 465 litres of sunflower oil, and 496 litres of soyabean oil. He wants to pack these into tins of equal capacity such that each tin is filled completely. Find the maximum capacity of such a tin.
3. If the polynomial $p(x) = x^4 - 6x^3 + 16x^2 - 25x + 10$ is divided by another polynomial $g(x) = x^2 - 2x + k$, the remainder comes out to be $x + a$. Find the values of k and a .

- Given that $x^2 + 1$ is a factor of the polynomial $p(x) = x^4 + x^3 + 8x^2 + ax + b$, find the values of a and b .
- If the sum of the first n terms of an arithmetic progression is given by $S_n = 3n^2 + n$, find the k^{th} term of this progression. Also, determine the common difference of the sequence.

Section D: Long Answer (4-5 marks each)

- Show that the square of any positive integer can be written in the form $4m$ or $4m + 1$ for some integer m . Using this result, prove that the square of any odd integer is of the form $8k + 1$ for some integer k .
- Two street light contractors, Company A and Company B, install lights on a straight road. Company A installs a light every 36 meters and Company B installs a light every 48 meters. If they both install a light at the starting point (position 0 meters), at what distance from the start will they next install a light at the same position? If the road is 1 kilometer long, how many times will their lights coincide (excluding the start)?
- If the polynomial $p(x) = x^4 - 6x^3 + 16x^2 - 25x + 10$ is divided by another polynomial $g(x) = x^2 - 2x + k$, the remainder is found to be of the form $x + a$. Find the values of k and a , and hence determine the quotient of the division.
- A company models the profit $P(x)$ (in thousands of rupees) of a product with respect to the production units x (in hundreds) as $P(x) = 2x^3 + x^2 - 5x + 2$. If the cost factor is represented by the divisor $g(x) = 2x - 1$, find the quotient $q(x)$ and verify the division algorithm $P(x) = g(x)q(x) + r(x)$ where $r(x)$ is the remainder.
- In an Arithmetic Progression, the sum of the first n terms is denoted by S_n . If the sum of the first 5 terms is 40 and the sum of the first 10 terms is 155, find the n^{th} term of the AP. Furthermore, determine the value of n for which the n^{th} term is 101.

Section E: Case Study (4 marks each)

- A packaging unit receives two consignments of items: 1260 units of item A and 1980 units of item B . The manager wants to pack these into identical cartons such that each carton contains only one type of item and the maximum possible number of items are packed in each carton, with no items leftover. Let H be the maximum number of items in each carton and L be the total number of cartons if all items were of type A packed in cartons of size H .
 - What is the value of H , the maximum number of items per carton?
 - (A) 90
 - (B) 120
 - (C) 180
 - (D) 360

- (ii) Calculate the value of L .
- (iii) If the number of cartons for item B is denoted by K , find the value of K .
- (iv) Which of the following is true regarding the decimal expansion of $\frac{1}{H}$?
- (A) Terminating
 - (B) Non-terminating repeating
 - (C) Non-terminating non-repeating
 - (D) None of these
2. In a school event, three batches of students are participating in a relay race. The lengths of the track segments assigned to the batches are 120 cm, 180 cm, and 300 cm respectively. The organizers want to mark the track with flags at equal intervals such that the distance between any two flags is the same and the largest possible distance is used. Let d be the distance between two flags.
- What is the value of d ?
- (A) 30 cm
 - (B) 60 cm
 - (C) 90 cm
 - (D) 120 cm
- How many flags are required for the 120 cm segment if flags are placed at both ends?
 - Find the total number of flags required for the 300 cm segment.
 - If a new segment of length x cm is added such that d remains the HCF of 120, 180, 300, and x , which of the following could be x ?
- (A) 70
 - (B) 80
 - (C) 420
 - (D) 100
3. A factory produces specialized machine components where the cost of production $C(x)$ (in thousands of rupees) for x units is given by $C(x) = x^4 - 6x^3 + 16x^2 - 25x + 10$. Due to a change in the manufacturing process, the cost function is being redesigned. The new process involves dividing $C(x)$ by a divisor polynomial $g(x) = x^2 - 2x + k$, which results in a remainder $R(x) = x + a$.
- If $x^2 - 2x + k$ is a factor of $C(x)$, what would be the remainder $R(x)$?
- (A) x
 - (B) 1
 - (C) 0
 - (D) $x + k$
- Perform the division of $x^4 - 6x^3 + 16x^2 - 25x + 10$ by $x^2 - 2x + 1$ to find the quotient.
 - Given the remainder is $x + a$ when divided by $x^2 - 2x + k$, find the value of k if $k = 5$ is a potential candidate. Calculate a for $k = 5$.

- (iv) If $p(x) = x^2 - 2x + k$ has equal roots, what is the value of k ?
4. An architect is designing a storage tank whose volume $V(x)$ is represented by a cubic polynomial $V(x) = 2x^3 + 3x^2 - 8x + 3$. The dimensions of the base are related to a linear factor of this polynomial. A safety regulation requires that the polynomial be verified using the division algorithm to ensure stability under different pressure conditions represented by a divisor $g(x) = x - 1$.
- (i) What is the remainder when $V(x)$ is divided by $(x - 1)$?
- (A) (A) 0
 (B) (B) 1
 (C) (C) -1
 (D) (D) 2
- (ii) Find the quotient when $V(x)$ is divided by $(x - 1)$.
- (iii) Factorize the quadratic quotient obtained in the previous step.
- (iv) State the three zeros of the polynomial $V(x)$.
5. A stadium seating arrangement is designed such that the first row has 20 seats, and each subsequent row has 4 more seats than the preceding row. The stadium has a total of 25 rows. Due to a renovation, the management decides to reconfigure the seating such that the n^{th} row contains a_n seats. Let S_n denote the total number of seats in the first n rows.
- (i) What is the number of seats in the 15^{th} row?
- (A) (A) 60
 (B) (B) 72
 (C) (C) 76
 (D) (D) 80
- (ii) Calculate the total number of seats in the stadium (all 25 rows).
- (iii) If a new section is added such that the sum of the first k rows of this section is given by $S_k = 2k^2 + 3k$, find the number of seats in the 10^{th} row of this new section.
- (A) (A) 39
 (B) (B) 41
 (C) (C) 43
 (D) (D) 45
- (iv) In the original stadium arrangement, if a row has 100 seats, which row number is it?